

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

**ATLAS: THESIS MANAGEMENT SYSTEM WITH AUTOMATED
RESEARCH DEFENSE SCHEDULING AND AI-ASSISTED
EVALUATION FOR THE COLLEGE OF ENGINEERING,
COMPUTER STUDIES, AND ARCHITECTURE
AT LYCEUM OF THE PHILIPPINES**

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Submitted to the Faculty of the
College of Engineering, Computer Studies and Architecture
Lyceum of the Philippines University-Cavite

In Partial Fulfillment
of the Requirements for the Degree of
Bachelor of Science in Computer Science
with specialization in Software Engineering

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July 2025

COLLEGE OF ENGINEERING, COMPUTER STUDIES, AND ARCHITECTURE

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

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ACCESS LEAF

**Bachelor of Science in Computer Science
with Specialization in Software Engineering**

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AT LYCEUM OF THE PHILIPPINES**

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This thesis entitled **ATLAS: THESIS MANAGEMENT SYSTEM WITH AUTOMATED RESEARCH DEFENSE SCHEDULING AND AI - ASSISTED EVALUATION FOR THE COLLEGE OF ENGINEERING, COMPUTER STUDIES, AND ARCHITECTURE AT LYCEUM OF THE PHILIPPINES** prepared and submitted by **WINSTON L. AGUSTIN, NEIL S. VICEDO, JERALD RYAN D. GERONA, and IVAN KERWIN R. ILANO**, in partial fulfillment of the requirements for the degree of **BACHELOR OF SCIENCE IN COMPUTER SCIENCE WITH SPECIALIZATION IN SOFTWARE ENGINEERING** is hereby accepted.

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STUDIES, AND ARCHITECTURE AT LYCEUM OF THE PHILIPPINES

We hereby declare that this thesis is our own work and that, to the best of our knowledge and belief, contains no material previously published or written by another person nor material to which to a substantial extent has been accepted for award of any other degree or diploma of a university or other institute of higher learning, except where due acknowledgment is made in the text.

We also declare that the intellectual content of this thesis is the product of our work, even though we may have received assistance from others on style, presentation, and language expression.

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LIST OF ABBREVIATIONS

LPU-C	Lyceum of the Philippines University Cavite
COECSA	College of Engineering, Computer Studies, and Architecture
LLM	Large Language Model
SQL	Structured Query Language
CMS	Content Management System
GA	Genetic Algorithm
ASS	Automated Scheduling System
DSS	Decision Support System
HTML	Hypertext Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
PHP	Hypertext Preprocessor
NLP	Natural Language Processing

ABSTRACT

Thesis defense management and evaluation processes within the academic institution are often challenged by scheduling conflicts, inconsistent assessments and limited tools to ensure originality in research titles. This study aims to develop ATLAS, a web-based system that utilizes AI to streamline scheduling, improve assessment accuracy, and promote academic integrity within the College of Engineering, Computer Studies, and Architecture (COECSA) at Lyceum of the Philippines University Cavite (LPU-C). ATLAS integrates the Genetic Algorithm for an optimized scheduling system and Gemini, a Large Language Model (LLM), to aid in plagiarism detection and sentiment analysis for thesis feedback. Data from student researchers, thesis advisers, and program chairs within COECSA were collected to evaluate system effectiveness based on ISO/IEC 25010 quality standards, covering functionality, usability, and reliability.

Keywords: thesis defense management, web-based system

CHAPTER I

INTRODUCTION

Background and Rationale of the Study

Thesis management systems play a crucial role in higher education by enhancing academic performance and supporting the timely completion of research projects. Digitalization is an effective method for enhancing processes and improving work efficiency, and this principle extends to the education sector for educators as well (Kauppinen, 2023). These systems are pivotal in integrating technology into academic workflows, improving communication, and ensuring high-quality output.

The integration of AI-driven tools into academic processes has become a critical solution for addressing the challenges faced by educational institutions, particularly in managing thesis-related tasks. Recent studies have highlighted key aspects of thesis management systems, such as scalability and maintainability, which enable institutions to handle growing student populations more efficiently (Khozaimi, 2023). Generative AI tools, such as Claude, have been shown to streamline the drafting and evaluation of final-year projects, improving both submissions and processes and faculty assessments (Pereira, 2023). Additionally, academic writing tools that provide real-time feedback have proven effective in enhancing students’ writing skills and increasing their chances of success (Storey,

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	2023). Furthermore, AI’s capability to analyze student performance data and offer personalized insights for evaluators enrich the thesis evaluation process, fostering tailored learning experiences and better academic outcomes (Abbas et al., 2023). These advancements highlight the transformative potential of AI in improving the efficiency and quality of thesis management systems. Moreover, studies reveal the ethical considerations of AI in academia. Tindle addresses concerns about academic integrity, urging institutions to establish guidelines for ethical AI use in assessments and writing (Tindle, 2023). Widodo examines AI’s application in lecturer performance evaluations, illustrating its capacity to process large datasets on teaching effectiveness, research, and feedback for informed decision-making Widodo (2023). Together, these studies underscore AI's transformative role in higher education, from improving student outcomes and academic processes to enhancing institutional decision-making, while emphasizing the need for ongoing research and ethical frameworks to ensure AI’s effective implementation. The College of Engineering, Computer Studies, and Architecture (COECSA) at Lyceum of the Philippines University-Cavite (LPU-C) is committed to cultivating students' practical expertise, promoting lifelong learning, and developing globally competent leaders. COECSA holds distinguished accreditations, such as Level 3 PACUCOA accreditation for the BS Information Technology program and Level 2 for its engineering programs. Furthermore, the college ensures industry readiness by offering professional certifications like MTA	
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	and CCNA, while producing graduates who exemplify the core values of truth, fortitude, and service (LPU, n.d.). Despite its academic excellence, COECSA faces logistical challenges in managing research presentations due to its growing student population and reliance on manual system. Scheduling defense, coordinating the availability of students, advisers, and panelists, and organizing documents are time-consuming tasks that often lead to conflicts, inefficiencies, and delays. Manual processes further complicate thesis management, as they are prone to errors in grading this hinders the tracking of schedules and submissions, and create administrative burdens, such as duplicate titles and unsecured data storage (Cagana & Istiono, 2024; Nugraha, 2022). These inefficiencies not only slow down communication and thesis progress but also highlight the need for a more streamlined and automated approach. As Kauppinen et al. (2023) emphasizes, digitalization in education can address these challenges by providing tools that enhance efficiency, improve organization, and generate valuable data for decision-making, ensuring a smoother thesis management process. The study proposes the development of ATLAS: Thesis Management System with Automated Research Defense Scheduling and AI-Assisted Evaluation, a comprehensive solution tailored for COECSA at Lyceum of the Philippines University-Cavite. ATLAS integrates a suite of AI-driven tools to address key challenges in thesis management. The Thesis Topic Decision Tool helps students select impactful research topics by analyzing current trends.	
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The Research Title Acceptance Tool, powered by Gemini Pro LLM for its advanced capabilities in multi-turn dialogues, context-aware interactions, and robust performance, making it an ideal choice for developing reliable, high-quality applications that integrate seamlessly (MindsDB, 2024). ensures originality by assessing the uniqueness of proposed titles. The Scheduling System, utilizing a Genetic Algorithm, automates defense coordination, optimizing availability and reducing conflicts. Additionally, the Content Management System (CMS) centralizes academic resources, streamlining administrative tasks. The Requirement Checker Tool tracks document submissions and sends reminders, while the Pass Recommendation System employs AI-driven sentiment analysis to provide consistent feedback during evaluations. Together, these tools aim to improve efficiency, accuracy, and transparency in thesis management, ultimately enhancing the academic experience for students and faculty alike.

Objectives of the Study

The general objective of this study is to develop ATLAS: Thesis Logistics and AI System to streamline the scheduling, evaluation, and management of thesis presentations for COECSA at Lyceum of the Philippines University – Cavite.

Specifically, the project will:

- 1. Develop ATLAS: Advance Thesis Logistics and AI System for LPU-Cavite COECSA Department with the use of the following features.
 - a) Thesis Topic Decision Tool

- b) Research Title Acceptance Tool
- c) Scheduling System
- d) Content Management System
- e) Requirement Checker Tool
- f) Pass Recommendation System

- 2. The project will use HTML/CSS and JavaScript for front-end flexibility and a user-friendly interface. PHP will be used for backend to manage server-side logic and real-time data processing. For database management, MySQL will be utilized to provide scalable and efficient data storage.
- 3. Evaluate and improve the system using ISO/IEC 25010 standards.

Significance of the Study

The development of ATLAS: Automated Research Scheduling Using Genetic Algorithm and Gemini Assisted Evaluation for LPU-Cavite COECSA Department aims to change how research presentations are scheduled, evaluated and managed. This study is important as it introduces an AI-driven system that rationalizes the logistical challenges faced by both students and school faculty in organizing thesis defenses, while improving the overall efficiency and transparency of the academic process. Student Researchers of LPU-C will

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	greatly benefit from this system. ATLAS offers students clear and organized scheduling, performance assessments, and presentation requirements. The system improves the research and defense processes accountability and transparency. Thesis and Capstone Advisers will also experience significant advantages. ATLAS will provide them an efficient platform for scheduling optimizing time management, research presentations, and minimizing conflicts. By using the system, advisers may monitor the students' area of improvement and make well informed decisions. Research Teachers of LPU-C COECSA will benefit from the system's AI capabilities in facilitating research title validation, plagiarism detection, and communication between students and faculty. By implementing the system, LPU-C's commitment on using modern technologies to improve academic processes. Future Researchers stand to benefit from this study as it can serve as a model for incorporating AI into academic scheduling and assessment systems. Future Researchers could utilize this framework to improve and innovate the academic scheduling and evaluation system. Scope and Limitation This study is centered on the development and implementation of ATLAS: Automated Research Scheduling Using Genetic Algorithm and Gemini Assisted Evaluation, specifically customized for the College of Engineering, Computer Studies and Architecture (COESCA) at Lyceum of the Philippines University-	
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	Cavite (LPU-C). The system includes the following key features such as Scheduling System which automates the scheduling of thesis defenses to minimize conflicts and improve efficiency, Pass Recommendation Tool that utilizes sentiment analysis via Gemini, to analyze feedback and suggest pass/fail outcomes, Research Title Acceptance that detects similarities between proposed and existing research titles to prevent plagiarism and ensure originality, Requirement Checker Tool that tracks thesis chapter submissions and ensures compliance with deadlines and academic requirements, and lastly Thesis Topic Decision Tool that provides students with topic suggestions based on trends in computing and AI. The research will take place at LPU-C and will concentrate on overcoming the logistical and academic challenges faced by COESCA, with data collection being carried out over one semester. The system will be applicable only within COESCA, with its features specifically designed to meet the needs of the department. These features may require modifications for use in other departments. The study will include key stakeholders such as students, research advisers, and panelists. It will not address broader administrative activities, plagiarism checks beyond title comparisons, or other academic programs at the university. Although AI-driven tools like the Gemini Pro LLM will aid in the decision-making process, human input is important for final grading and evaluation to maintain fairness and accuracy. The system necessitates an internet connection and compatible devices, and it is built for web-based access.	
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	ATLAS requires a reliable internet connection, and it requires a Windows 10 operating system version 10.0.19045 Build 19045. For CPU a minimum of Intel i3 3rd Gen or Ryzen 3 3000 series or 4000 series APU and a minimum of 4GB RAM. As for the resolution, a minimum of resolution of 720 x 565 (also with a 16:9 aspect ratio) and 720 x480 (for 4:3). For browsers, Google Chrome 131.0.6778.109 (Official Build) (64-bit), Opera version LVL6 (core: 114.0.5282.233), Microsoft Edge version 131.0.2903.86 (Official build) (64-bit), and Firefox version 131.0.2 (64-bit). For mobile users, the minimum Android Version is Android 10, for iOS is iOS 17. As for the resolution for mobile devices, the system requires a minimum resolution of 565 x 720 (with a 16:9 aspect ratio) and 720 x480 (for 4:3)to ensure optimal functionality and a fully responsive design. These specifications allow users to effectively use all features, with a layout designed for both mobile and desktop environments.	
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CHAPTER II

REVIEW OF RELATED LITERATURE

The proponents of this project collected relevant literature and studies to facilitate the research. This collection aims to assist readers in understanding the study's content. The literature review clarifies terms closely linked to the research, while the review of related studies delves into findings and recommendations important to this project. Also, a synthesis is provided at the end of each topic to show the connection between the literature and the researcher's work.

Genetic Algorithm

Genetic Algorithms are powerful optimization techniques inspired by the principles of natural selection and genetics. Since their inception by Holland in the 1970s, GAs have evolved into versatile tools widely applied across various domains, including engineering, artificial intelligence, and computational biology. The fundamental mechanism of Genetic Algorithm involves iteratively searching for optimal solutions through processes analogous to biological evolution, such as selection, crossover, and mutation (Chen and Okle, 2024). Genetic Algorithms have garnered significant attention in recent years due to their versatility and effectiveness in solving complex optimization problems across various fields. Their

improves on this by integrating AI responsibly. Through tools like the Thesis Topic Decision Tool and Research Title Acceptance Tool, the system supports students in selecting relevant and unique research topics while ensuring originality. Additionally, the Pass Recommendation System ensures that AI aids decision-making without replacing critical human evaluation, safeguarding both academic integrity and learning outcomes. (Leary ,2023)

LLM such as the Gemini Pro LLM used in ATLAS, adds a layer of AI-driven intelligence for thesis evaluations and feedback. LLMs can analyze text data, such as research titles or student-submitted documents, to provide relevant feedback, detect potential issues, and suggest improvements. For instance, the LLM helps in assessing the uniqueness of thesis titles by comparing them with existing works and generating recommendations to avoid redundancy or plagiarism. The use of LLM in ATLAS ensures that evaluations are more consistent and precise, as it supports the decision-making process with AI-driven insights, while human reviewers still maintain control over the final evaluations.

Thesis Management System

The development of thesis management systems has become increasingly relevant in the context of higher education. The integration of information technology into academic processes has been shown to enhance the quality and efficiency of thesis supervision and completion. Tiwow (2023) emphasizes that aligning thesis projects with the actual needs of the IT industry can make student

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	research more relevant and impactful. By implementing effective time management programs and project management training, students can approach their thesis projects with greater confidence and depth. This technological integration not only aids in project execution but also fosters a more engaging learning environment. In addition, the introduction of specialized thesis management systems, such as Penamasy, has been proposed to streamline the thesis writing process. Semma (2023) presents this system as a solution to the significant challenges students face during thesis writing, which often leads to stress and confusion. By providing structured support and resources, such systems can alleviate some of the burdens associated with thesis completion. Furthermore, a study by Khozaimi explores the development of a backend application for a thesis management system. This research illustrates how adopting a microservice approach can significantly enhance the scalability and maintainability of thesis management systems (Khozaimi, 2023). By allowing for modular updates and improvements, such systems can adapt to the evolving needs of academic institutions without disrupting ongoing processes. The study underscores the importance of technological innovation in thesis management, suggesting that such advancements can lead to improved service delivery and better management of academic activities. This is particularly relevant in the context of increasing student numbers and the corresponding demand for efficient academic support systems.	
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	The literature highlights the critical role of thesis management systems in enhancing academic performance. The integration of technology is essential for fostering a positive environment for thesis completion. As institutions continue to navigate the complexities of modern education, the development and implementation of effective thesis management systems will be pivotal in supporting students' academic journeys. Automated Scheduling System The development of automated scheduling systems has gained significant attention in recent years, particularly with advancements in machine learning and artificial intelligence. The integration of Automated Scheduling Systems (ASS) in educational institutions has gained significant traction in recent years, driven by the need for efficiency and effective resource management. These systems leverage advanced algorithms and artificial intelligence (AI) to streamline the scheduling process, thereby alleviating the administrative burden on educators and enhancing the overall educational experience for students. One of the primary advantages of Automated Scheduling Systems is their ability to optimize class schedules based on various constraints such as classroom availability, instructor schedules, and student preferences. For instance, the development of an automated scheduling system using the 1C:Enterprise platform has demonstrated significant improvements in managing educational processes within higher education institutions. This system not only reduces the time required for scheduling but also	
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	facilitates easier modifications to the timetable, ensuring that the educational process runs smoothly (Rymar & Voytko, 2021). Similarly, research has shown that employing genetic algorithms can effectively address complex scheduling problems in universities, allowing for the generation of optimal course schedules that meet diverse constraints (Lei et al, 2024). In addressing the challenges of scheduling in universities, (Chen et al., 2021) focused on optimizing class schedules through automated conflict resolution and efficient resource management. Their work highlights the importance of minimizing scheduling conflicts to enhance the educational experience for both students and faculty. While their study employed traditional software algorithms to achieve these goals, ATLAS applies AI-driven scheduling algorithms to further enhance the accuracy and efficiency of the scheduling process. This approach not only streamlines the scheduling of research presentations but also optimizes panelist and student availability, offering a more flexible and responsive solution to scheduling challenges. One notable approach to automated scheduling is the use of mixed-integer programming (MIP) for university course timetabling. Rappos et al. present a comprehensive MIP model that addresses the complexities of scheduling courses while considering various constraints such as classroom availability and instructor assignments. Their findings indicate that MIP can effectively minimize conflicts and optimize resource allocation, making it a suitable choice for educational institutions facing intricate scheduling challenges (Rappos et al., 2022).	
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	Additionally, the integration of genetic algorithms (GAs) into scheduling systems has been explored by Dogan and Yurtsal, who developed a decision support system for exam scheduling. Their system utilizes GAs to efficiently assign courses to time slots, demonstrating the algorithm's capability to handle large datasets and complex constraints (Dogan & Yurtsal, 2021). Another innovative approach is highlighted in the work of Uma et al., who developed an automatic timetable generation system that incorporates artificial intelligence to optimize resource utilization. This system takes inputs such as subjects, teachers, and workloads to generate timetables that adhere to predefined constraints, significantly reducing the manual effort required in traditional scheduling processes (Uma et al., 2023). The architecture of this system emphasizes adaptability and efficiency, showcasing how AI can enhance automated scheduling. Automated Scheduling System is another critical component that simplifies the logistics of research presentations and defenses. By leveraging AI and algorithms, this system automatically aligns the schedules of students, advisers, and panelists, ensuring that conflicts are minimized and that presentations can proceed smoothly. It considers various constraints, such as availability, deadlines, and institutional requirements, to generate the most efficient and conflict-free schedule possible. This automated process reduces the manual workload on administrators and minimizes errors, making it easier to manage multiple research defenses in a growing academic environment.	
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Decision Support System

Decision Support System is a computer program that helps decisions, assessments, and actions inside a company or organization. Massive data sets are sorted through and analyzed by a Decision Support System, which then compiles detailed information that may be utilized in decision-making and problem-solving. The theory and practice of decision support systems are progressing and occasionally intersecting with other notable breakthroughs in information technology, including organizational computing, e-commerce, and pervasive computing (Abdul-kareem et al., 2023).

Decision Support System is essential in guiding students and faculty members throughout the thesis process by providing structured recommendations based on analyzed data. In the context of ATLAS, the DSS helps students select thesis topics that align with current academic trends and their strengths. By evaluating available data and research trends, the DSS supports more informed decision-making, ensuring that students work on relevant and impactful topics. The system uses this decision support mechanism to provide objective recommendations, which helps streamline the thesis selection and development process, improving both the quality and relevance of the research produced.

Structured Query Language

Structured Query Language (SQL) remains a cornerstone in the management and retrieval of data within relational database management systems

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	proposed an improved CMS framework that leverages cloud-based tools to centralize data and streamline operations, ultimately enhancing student success. Building on this, our thesis aims to integrate AI-driven technologies, such as large language models (LLMs), to further optimize not just administrative tasks but also academic decision-making, research topic selection, and scheduling, providing a more dynamic and personalized solution. The LPU-C-DCS faced challenges with manually managing the scheduling, grading, and administration of capstone and research presentations due to a growing student population and faculty workload. To address this, (Leaño et al, 2013) created an online system that automated scheduling, grading, and report generation, reducing errors and saving time. Our system improves on this by incorporating AI-driven scheduling, LLM-powered feedback analysis, automated grading support, and a more robust content management system. These enhancements will further streamline the process, reduce conflicts, and provide intelligent assistance for both faculty and students CMS will serve as a crucial tool for managing academic content within ATLAS. It provides a flexible platform for faculty members to easily update rubrics, research titles, team lists, and panelist assignments without needing technical expertise. The CMS ensures that all important documents and data related to thesis projects are well-organized and easily accessible. By centralizing content management, the CMS will help streamline academic processes, reduce manual	
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Artificial Intelligence in Education

AI-powered adaptive learning platforms have shown promising results in improving student performance, particularly in subjects like mathematics. For instance, a study demonstrated that these platforms positively influenced students' mathematical skills, indicating that AI can effectively tailor educational experiences to individual learning needs (Dabingaya, 2022). This aligns with findings from other research that emphasizes the role of AI in personalizing learning pathways and providing real-time feedback, which can significantly enhance student engagement and achievement (Trang, 2024).

The application of Artificial Intelligence (AI) in research has gained significant traction in recent years, particularly in the context of academic writing, data analysis, and educational methodologies. This literature review synthesizes recent studies that highlight the transformative potential of AI across various research domains, emphasizing its role in enhancing academic performance, improving research methodologies, and addressing ethical considerations. AI's integration into academic writing has been a focal point of recent research. For instance, this discusses how AI is reshaping academic research, particularly in personalized learning and intelligent tutoring systems, which bridge the gap between academic results and industry needs (William, 2024). This sentiment matches with (Pigola, 2023), who notes that AI's influence extends beyond mere data processing to fundamentally altering the landscape of academic research itself. The potential of AI tools, such as Gemini, to assist in literature reviews and data

Conceptual Framework of the Study

The proponents undertaking the project must acquire knowledge in HTML, CSS, JS, SQL, and PHP, as these skills are essential for the development of Decision Support System. The framework shows the system’s components and their relationship, focusing on how an AI-driven website processes can optimize scheduling and assessment within the academic environment.

As illustrated in Figure 1 below, the Scrum model was chosen for this study because it provides an agile framework that enables iterative development, continuous feedback, and adaptability, ensuring the efficient implementation of ATLAS’s complex, user-driven features.

Figure 1. *Conceptual Framework of ATLAS.*

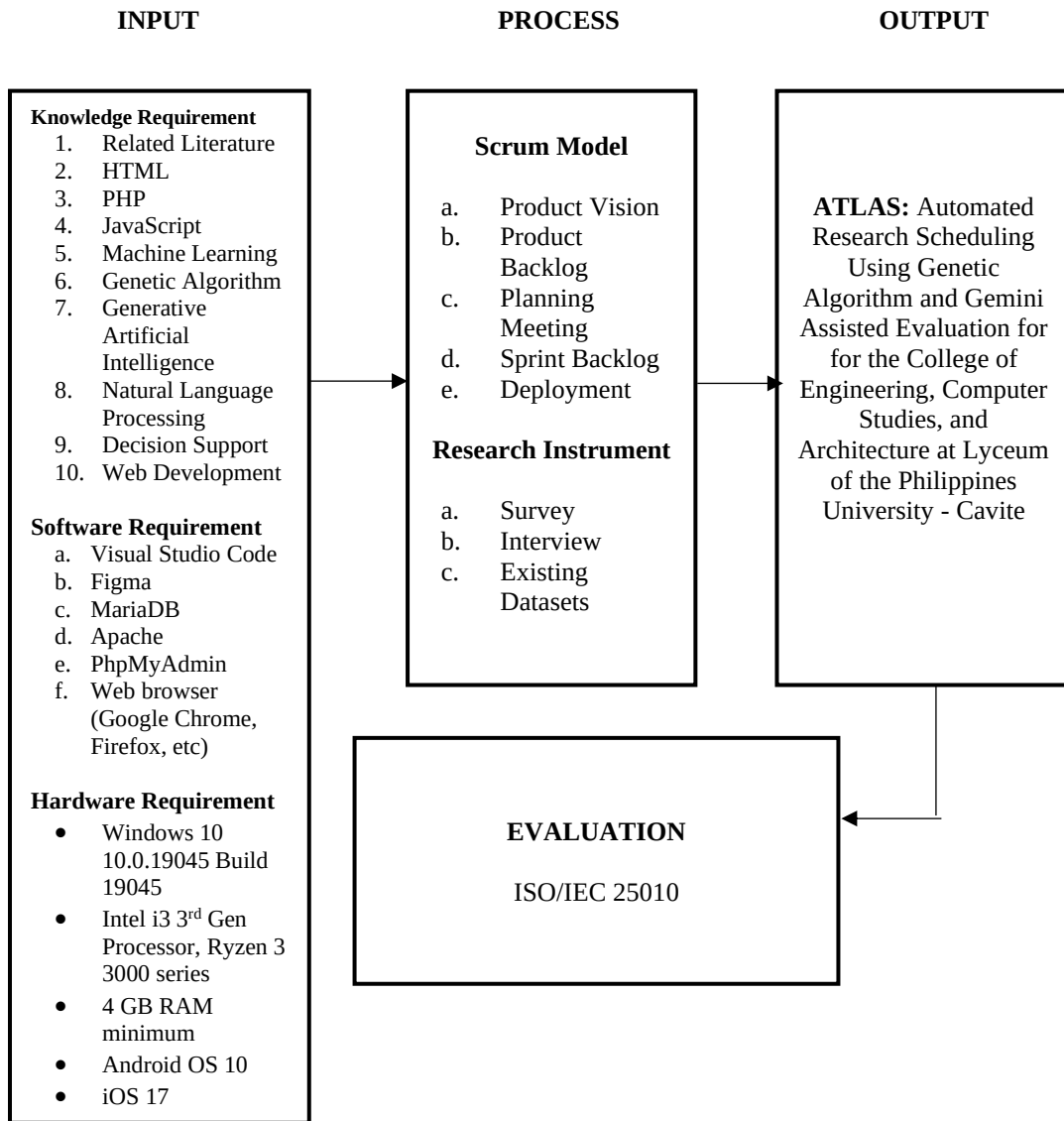


Figure 1 illustrates the project's framework representing the Input, Process, and Output phases of the study. The Input section shows the required Knowledge requirements, Software requirement, and Hardware requirements which are necessary for the proponents to develop the system. Next is the Process section,

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	Content Management System. A digital platform that helps users create, manage, and publish content for websites and apps. In ATLAS it streamlines document management, ensures transparency, reduces redundancy, and minimizes errors, providing students and faculty with easy access to essential materials for a cohesive academic experience. Gemini. A Large Language Model (LLM) developed by Google AI. In ATLAS, Gemini is used as it empowers the Thesis Topic Decision Tool, Research Title Acceptance, and Pass Recommendation features. Large Language Model. A type of AI program that uses deep learning to process and generate human language. LLM allows the features in ATLAS to work. Genetic Algorithm. A computational method that uses natural selection and genetic operators to solve problems by evolving a population of candidate solutions. In ATLAS, GA optimizes scheduling by efficiently assigning research presentations to available slots. MariaDB. A free, open-source relational database management system that is used for storing data in a tabular format. It is used in ATLAS as a server database.	
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PHP. A server-side scripting language embedded in HTML in its simplest form. PHP is used where the server-side is coded.

Apache. A free, open-source web server that delivers web content over the internet. It is used in ATLAS as the web server

Visual Studio Code. A source code editor that helps developers build, debug, and test code. VS Code is used as the IDE for ATLAS

Decision Support System (DSS). An AI-based system that aids in academic decision-making by analyzing data to recommend thesis topics for students and assess work quality for advisers. It ensures informed, unbiased, and consistent recommendations.

Scheduling System. Uses a Genetic Algorithm to optimize thesis defense schedules by aligning participant availability and institutional constraints. It minimizes conflicts, reduces manual effort, and improves efficiency.

Thesis Topic Decision Tool. An AI-powered tool that analyzes research trends to suggest unique, relevant, and impactful thesis topics aligned with academic and industry standards.

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Requirement Checker Tool. Tracks mandatory thesis defense documents, alerting students to incomplete or missing items to ensure timely completion and a smooth defense process.

Pass Recommendation System. Uses Gemini Pro LLM to analyze panelists' feedback and scores, providing preliminary AI-based pass/revision recommendations. It reduces grading inconsistencies and errors while maintaining academic rigor.

Research Title Acceptance Tool. Assesses title originality by comparing proposed titles with online research databases to prevent duplication and ensure unique contributions.

CHAPTER III

METHODOLOGY

Research Design

In this study, the proponents used a quantitative survey research approach to evaluate and see the potential effectiveness of the Atlas: Thesis Management System with Automated Research Defense Scheduling And AI-Assisted Evaluation For The College Of Engineering, Computer Studies, And Architecture At Lyceum Of The Philippines. This approach was chosen to get valuable data from students, advisers, and program chairs. The data will be analyzed to see and understand the relevance of the proposed system.

To optimize scheduling, the Genetic Algorithm was chosen as the main algorithm for ATLAS. Genetic Algorithm is a robust optimization technique inspired by the process of natural selection. It is used in ATLAS to efficiently allocate limited slots for research presentations and to avoid conflicts, ensuring that all users of ATLAS have an optimized and organized schedule.

Additionally, Gemini, a Large Language Model (LLM), is utilized within ATLAS to perform tasks related to natural language processing (NLP). Gemini aids in the text analysis of ATLAS, especially in sentiment analysis of feedback from

that, compared to other algorithms, the GA yields better average fitness values and computational efficiency, making it an ideal choice for addressing the multi-objective demands of course scheduling. (Xu., 2021)

Gemini, with its dual encoder model and memory-augmented mechanism, is uniquely positioned to excel in thesis management compared to other AI models. Unlike traditional algorithms, which may lack the ability to handle complex, multi-turn dialogues, Gemini’s architecture is specifically designed to understand and engage in sequential, context-rich interactions. This enables it to navigate the nuanced and evolving steps in thesis tasks, such as suggesting relevant topics, validating research titles, and providing pass recommendations. The model’s memory-augmented mechanism allows it to retain context across multiple dialogue turns, making it ideal for tasks that require consistency and continuity, such as following up on a student’s thesis development over time. Additionally, Gemini’s fine-grained dialogue state tracking ensures accurate and relevant responses by keeping track of the evolving context, which is essential in academic environments where clarity and adherence to specific guidelines are critical. While other Large Language Models (LLM) offers strengths in open-ended conversation and task-specific support, Gemini’s tailored approach to multi-step dialogue and task tracking makes it a superior choice for the specialized requirements of thesis management. (Krify, 2024)

The proponents chose Gemini over GPT because it meets the specific technical requirements of our project, particularly in handling multi-turn dialogues

and providing context-aware interactions tailored for academic processes. Google’s Gemini is highlighted as one of the best models for developing robust applications due to its combination of high performance, versatility, and cutting-edge features.

As a proprietary model from one of the leading AI companies, Gemini benefits from Google's extensive resources, data, and expertise in natural language processing and machine learning. It excels in tasks requiring deep reasoning and a broad general knowledge base, making it suitable for a wide range of applications. Additionally, Gemini's integration with Google's ecosystem and tools enhances its utility for developers, offering seamless compatibility with cloud services, APIs, and advanced deployment options. For applications needing top-tier performance out of the box, Gemini is presented as a strong choice, alongside other models particularly when quality and reliability are top priorities. (MindsDB, 2024) Also, in an article by (Jackson, 2023) it is stated that Google AI is one of the ethical companies that practices AI. Gemini is identified as the most adversarially robust AI model on the leaderboard due to its superior performance in resisting adversarial attacks designed to exploit vulnerabilities in AI systems. This robustness demonstrates its advanced architecture and training methodology, enabling it to maintain accuracy and reliability under challenging conditions (Scale, 2024).

Stratified random sampling will be employed to gather responses from Professors, Research Advisers, Research Panelists, Department Chairs, and Students through an online survey. This approach involves dividing the population into distinct subgroups (strata) based on their roles within the thesis process. A

random sample will then be drawn from each subgroup, ensuring a proportionate and representative selection of participants. By capturing input from all relevant groups, this method ensures comprehensive feedback on the college's thesis system and the potential benefits of an AI-driven decision support tool, while minimizing bias and enhancing the validity of the research findings.

Participants of the Study

The participant of this study includes three key groups from LPU-C COECSA Department: students, thesis advisers/professors, and the program chair. All the mentioned participants are vital in giving valuable feedback on potential improvements of ATLAS.

- Student Researcher/Proponent will provide insights on how the system supports their scheduling, performance tracking and research title management
- Research advisor monitoring the students' progress, and improve the evaluation of research presentations
- Research Panel will assess the student's performance during defense.
- The Program Chair will offer its insights into the system's overall impact on the department.

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	Research Locale This study will be conducted at Lyceum of the Philippines University – Cavite (LPU-C), specifically within the College of Engineering, Computer Studies, and Architecture (COECSA), located in General Trias, Cavite. LPU-C is known for its commitment to academic excellence and fostering innovation, making it an ideal setting for this study. COECSA Department has been chosen as it houses more complex research presentations and academic assessments. The department is home to a diverse group of faculty members and students who actively engage in academic research, making them the main beneficiary of the proposed system. By focusing on COECSA, the study aims to address real-world challenges in managing research defenses and enhance the efficiency of academic assessments. Research Instrument To gather data for this research, an online survey questionnaire using Microsoft Forms. The survey will target the key stakeholders including Professors, Research Advisers, Research Panelists, Department Chairs, and Students. The objective is to gain valuable insights into the current thesis system used within the college and to explore the potential of implementing an AI-driven decision support tool.	
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Data Gathering Procedure

The proponents relied on online surveys with the selected respondents. The survey was sent through email and can be accessed by using their LPU account.

The questions for the survey were made based on the research objectives. The survey has categories depending on the respondents if either they are students, thesis professors/advisers and program chair. The respondents' privacy and confidentiality were strictly maintained throughout the data gathering process.

Multiple Constraints Analysis

The ATLAS project is influenced by several constraints that shape its development and deployment. These constraints are primarily technical, user-related, and regulatory. While other areas, such as environmental and health-related constraints, have no relevance to this system.

First technical constraints include the system's dependency on hardware such as internet access, mobile devices and computers, which are critical for accessing and operating the system. Limited access to these resources, especially in areas with poor connectivity or insufficient hardware availability, could impact the system's functionality and user experience. A stable internet connection is essential, as the system cannot function or be accessed without it, potentially delaying research processes.

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	Regarding data privacy constraints, the system handles sensitive information such as student, adviser, and administrative accounts, necessitating strict measures to protect this data. All personal information will be anonymized to ensure confidentiality, and access will be restricted to authorized personnel only, reducing the risk of unauthorized access or data breaches. These measures align with privacy laws and reinforce user trust in the system's integrity and security. As for the constraints that are not applicable to this project, environmental constraints are not relevant since ATLAS is a software-based project, it has no direct interaction with physical environments or ecosystems. It uses existing digital infrastructure, resulting in no significant ecological impact. Similarly, health and safety constraints are not relevant, since the project is entirely digital, it does not involve physical risks to users. There are no requirements for emergency preparedness measures or occupational health precautions. Lastly, sustainability constraints, while sustainability in terms of long-term system usability is a consideration, there are no concerns regarding physical resource management or ecological sustainability, as the system does not rely on consumables	
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System Development Process

The proponents utilized the Scrum model for the development of ATLAS: Automated Research Scheduling Using Genetic Algorithm and Gemini Assisted Evaluation for LPU-C. The Scrum model has 5 components that the proponents utilized. This includes Product Vision, Product Backlog, Planning Meeting, Sprint Backlog, and Deployment.

Figure 2. Scrum Model

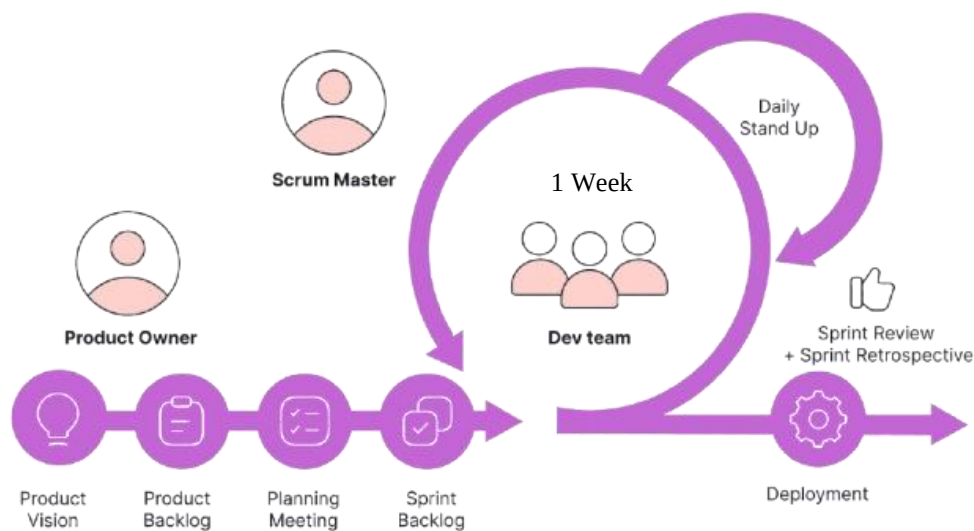


Figure 2 illustrates the process of Scrum Model. Below discusses each step pertaining to the actions of the life cycle:

Scrum Framework Overview

The Scrum model operates in short sprints typically lasting between 1-4 weeks. But for this study the researchers intend to operate weekly. Each sprint consists of planning and development. Key roles in Scrum includes Product Owner, Scrum Master, and Development Team, each roles plays a crucial role ensuring the success of the project.

Roles:

Product Owner: Defines the product vision, prioritizes the product backlog, and ensures value delivery.

Scrum Master: Facilitates Scrum processes, removes blockers, and drives team improvement.

Development Team: Builds, tests, and documents the product increment collaboratively.

Artifacts and Events:

Product Vision: The goal and purpose of the product.

Product Backlog: A prioritized list of all tasks, features, and requirements for the product.

Sprint Planning Meeting: A session to define the sprint goal and select backlog items for the sprint.

Sprint Backlog: A focused subset of the product backlog for the sprint.

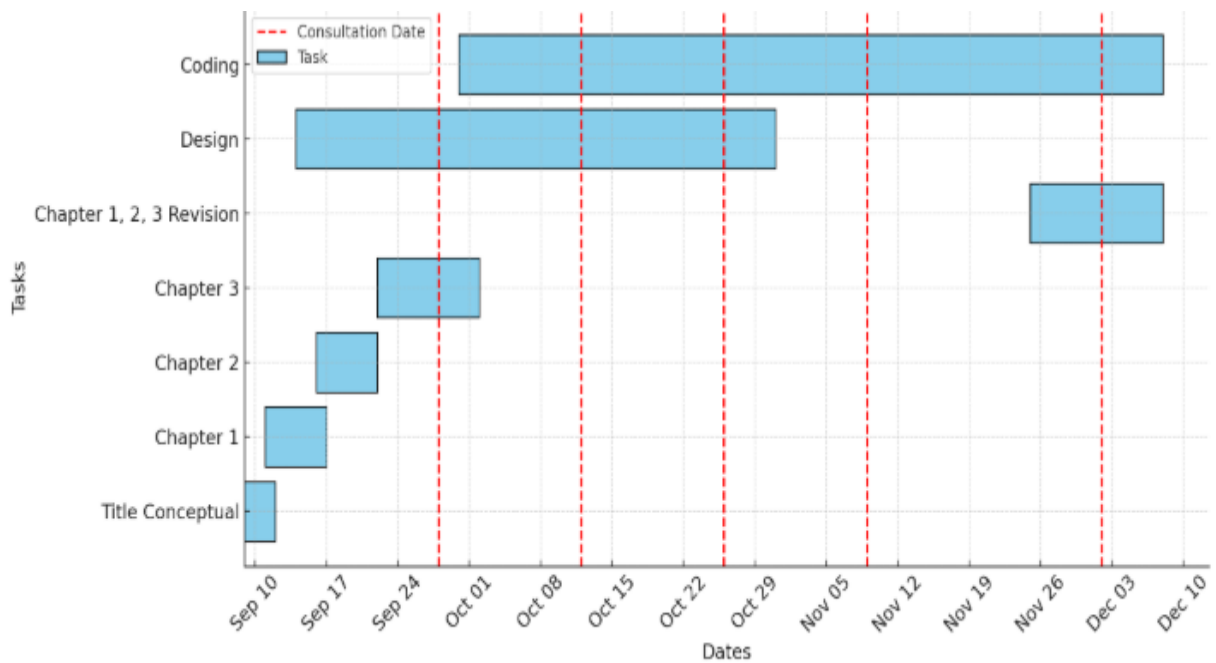
Daily Standup: A quick meeting to align team progress and identify blockers.

Sprint Review: A presentation of completed work for stakeholder feedback.

Sprint Retrospective: A reflection meeting to identify and plan process improvements.

Deployment: The release of the product increment to stakeholders or production.

Figure 3. Gantt Chart of ATLAS



The Gantt Chart of ATLAS outlines the timeline for the various tasks involved in the project's development, providing a clear breakdown of activities, durations, and consultation dates. The tasks are organized from conceptualization

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	to revision phases. Starting on September 10 and ending on December 10. Key activities include Title Conceptualization, Chapter 1-3 Writing, Design, and Coding, with tasks arranged sequentially based on dependencies and priority. Chapter 1, Chapter 2, and Chapter 3 were completed progressively between mid-September and early October. Following this, the Design phase spanned from mid-September to late October, overlapping with the start of the Coding process, which is the longest-running task from October 1 to December 10. Additionally, the Gantt chart highlights specific Consultation Dates (marked in red dashed lines) occurring on October 1, October 29, November 5, and November 26. These consultations are strategically placed to align with major task milestones, ensuring feedback is received during critical phases such as design completion and coding progression. The Chapter 1,2,3 Revision task occurs toward the project's final phase, between late November and early December. This well-structured timeline ensures a balance of development, feedback and revision, critical for the efficient and organized completion of ATLAS.	
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Figure 4. Base Genetic Algorithm

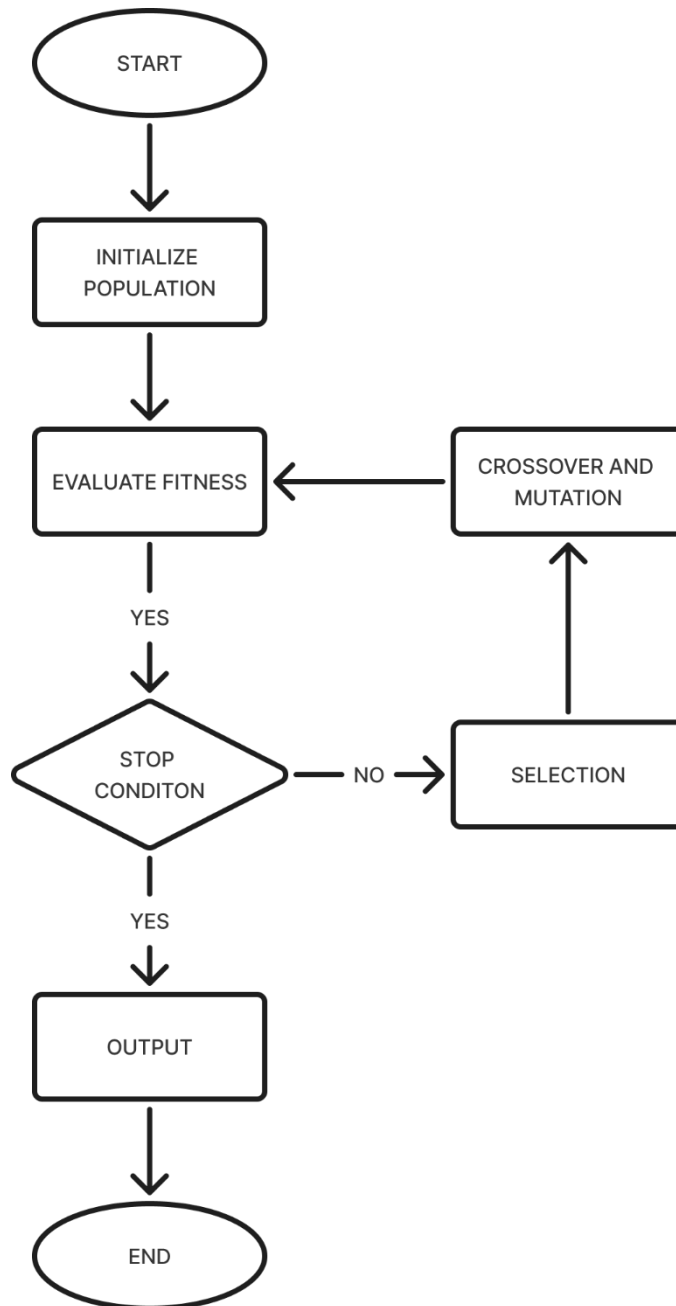
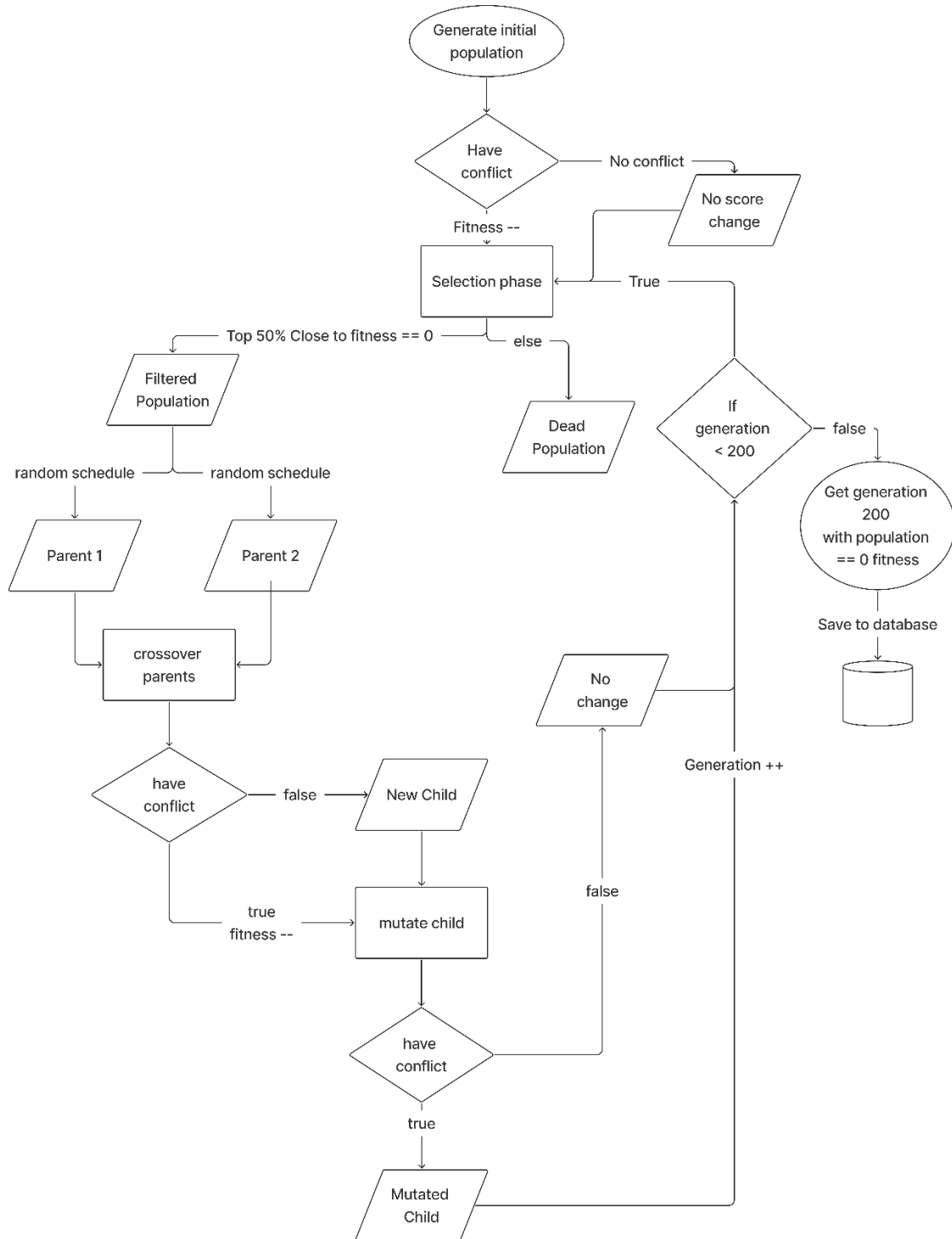


Figure 5. Genetic Algorithm Simulation



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Parameters:

- **Teams:** Team A and Team B
- **Panelists:** P1, P2, P3, and P4
- **Rooms:** Room 1, Room 2
- **Time Slots:** 9:00 AM, 10:00 AM
- **Days:** December 11, 2024

Initial Population:

- **Population Size:** 2
- **Generations:** 2 (example)
- **Mutation Rate:** 10% (0.1)

Initial Population

Table 1. Individual 1 (Schedule)

Team	Panelists	Room	Time Slot	Day
Team A	P1, P2, P3	Room 1	9:00 AM	December 11, 2024
Team B	P2, P3, P4	Room 1	9:00 AM	December 11, 2024

The table showed both teams having a schedule in the same room with a panelists P2 and P3 having overlapping schedules reducing fitness score by 20, 10 for each conflict.

Table 2. Individual 2 (Schedule)

Team	Panelists	Room	Time Slot	Day
Team A	P1, P2, P4	Room 2	9:00 AM	December 11, 2024
Team B	P3, P4, P1	Room 1	10:00 AM	December 11, 2024

The table shows a conflict panelist P4 is double-booked between the two teams A and B which deducts its fitness score by 10 compared to Individual 1, and there is a crossover between Parent 1 Individual 1 and Parent 2 Individual 2.

Table 3. Child (New Schedule):

Team	Panelists	Room	Time Slot	Day
Team A	P1, P2, P3	Room 1	9:00 AM	December 11, 2024
Team B	P3, P4, P1	Room 1	10:00 AM	December 11, 2024

Table 3 shows that panelist P3 has overlapping schedule and Room 1 has been booked twice where and that there is a mutation chance of 10% to change the team's schedule. Team B changes its room as part of the Mutation.

Table 4. Mutated Child

Team	Panelists	Room	Time Slot	Day
Team A	P1, P2, P3	Room 1	9:00 AM	December 11, 2024
Team B	P3, P4, P1	Room 2	10:00 AM	December 11, 2024

Table 4 shows no conflict after the mutation where the child of the 1st Generation has a fitness of 0 which is dominates and is better compared to Individual 2 with a fitness of -10. Another crossover may occur, but results will favor conflict-free schedules.

It will iterate depending on the set number of generation, the tuning of 100 population, 200 generations, and 0.1 mutation rate returned significant changes with least errors.

The input values will be sent as POST data to the run_scheduler.php file, which will also retrieve data from the database, including information about teams, users (panelists), and user schedules.

The algorithm will generate a randomized initial population of 100 solutions and iterate through 200 generations, using a learning rate of 0.1. Each team will be limited to a single schedule.

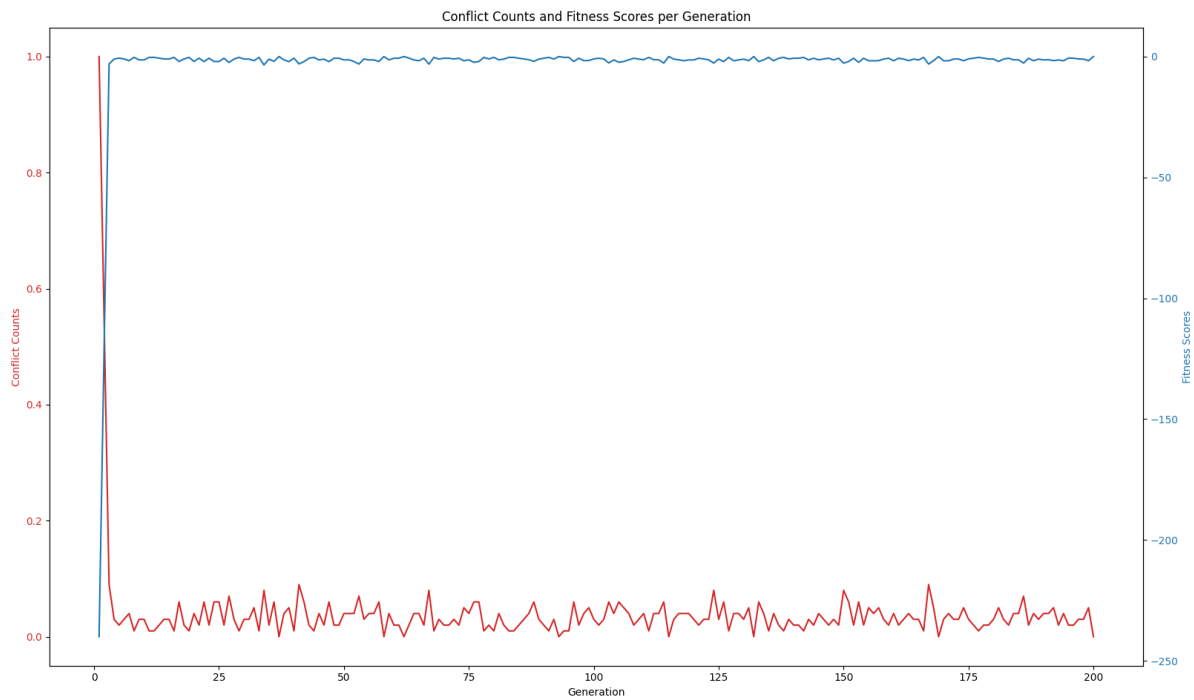
For each generation:

1. The algorithm computes the fitness score and counts conflicts for each individual in the population.
2. It calculates the average fitness and conflict count for the generation.
3. The solution with the least conflicts is selected, and genetic crossover is performed on its attributes (e.g., time slots, days, and rooms). These attributes are pooled and passed down to the next population.

The genes are then randomly assigned to the new population. If a newly generated gene assignment introduces more conflicts, the program reverts the changes and moves on to the next population member.

Finally, the population is sorted from the best to worst fitness scores, with the top individual representing the current best solution. Below is an example run.

Figure 6. Conflict Counts and Fitness Scores per Generation



The graph shows the performance of a genetic algorithm (GA) over generations, with two metrics plotted:

1. **Conflict Counts (left y-axis):** This likely represents the number of violations or conflicts (e.g., constraints in a scheduling problem) in the solution at each generation.

2. **Fitness Scores (right y-axis):** This represents how optimal or "fit" the solutions are (lower fitness scores could indicate better performance if minimizing is the goal).

Here's why the graph looks like this:

Initial Generations

- **Sharp drop in conflict counts:** In the early generations, the algorithm explores the search space and rapidly improves the population by eliminating many conflicts. This is typical as random solutions with high conflicts are refined through crossover, mutation, and selection.
- **Corresponding fitness improvement:** Fitness scores also improve quickly during these initial iterations, reflecting better solutions with fewer conflicts.

Stabilization Phase

- **Conflict counts oscillation:** After the initial rapid improvement, the red line stabilizes around a low value with slight fluctuations. This suggests the algorithm has reduced most conflicts but is struggling with local optimization and minor adjustments.
- **Fitness scores plateau:** Similarly, the blue line plateaus, indicating the population is converging to solutions that are near-optimal.

Reason for Oscillations

- **Genetic diversity:** Fluctuations in both metrics may result from mutations or crossovers introducing diversity into the population. This can momentarily worsen solutions before re-converging to better solutions.
- **Local optima:** The GA might be oscillating around a local optimum if global exploration is limited.

Observations

- The fitness scores and conflict counts don't seem to improve significantly after a certain point, which may indicate:
 - The algorithm has converged to a near-optimal solution.
 - Further refinement may require a hybrid approach, such as local search methods post-GA.

Mathematical Model of Genetic Algorithm

The mathematical model of Genetic Algorithm consists of several components that outline its functioning for solving optimization problems.

1. Population Representation

The population at generation t is denoted as $P(t) = \{x_1, x_2, \dots, x_n\}$, where x_1 represents an individual solution (a potential candidate). Each individual is encoded, often as binary strings, real values, or other representations depending on the problem.

2. Fitness Function

The fitness function evaluates the quality of each solution X in the population. Solutions with higher fitness values are considered better candidates for the problem

$$f: X \rightarrow \mathbb{R}$$

3. Selection Probability

Individuals are selected based on their fitness. The selection probability of an individual x_i is calculated as:

$$p(x_i) = f(x_i) / \sum_j f(x_j)$$

4. Crossover Operator

Crossover combines the genetic material of two parent solutions $x_p^{(1)}$ and $x_p^{(2)}$ to produce an offspring x_o :

$$x_o = \text{Crossover}(x_p^{(1)}, x_p^{(2)})$$

Common techniques include single-point, multi-point, and uniform crossover.

5. Mutation Operator

Mutation introduces randomness into the offspring to avoid local optima and maintain genetic diversity. It modifies an individual x_i with a small probability, producing x'_i :

$$x'_i = M(x_i)$$

For binary encoding, this might involve flipping bits; for real-valued encoding adding small perturbations.

6. Replacement Strategy

The next generation $P(t+1)$ is created by replacing some or all the old population $P(t)$ with offspring $O(t)$. The replacement can be generational (entire population replaced) or elitist (best individuals are retained).

$$P(t+1) = \text{Replacement}(P(t), O(t))$$

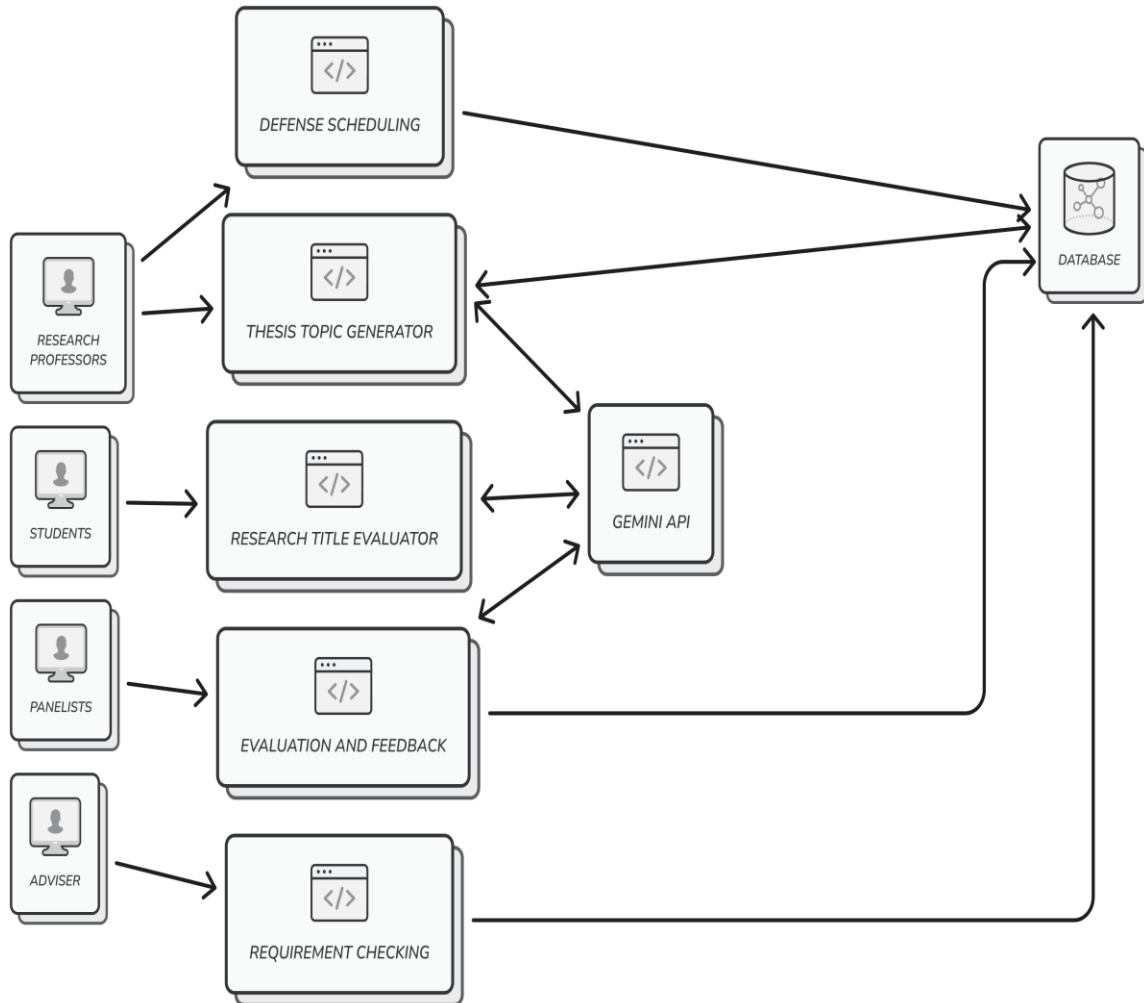
7. Termination Condition

The algorithm stops when one of the following conditions is met:

- A maximum number of generations is reached $t \geq T$.
- A satisfactory fitness value is achieved:

$$\exists x \in P(t) \text{ such that } f(x) \geq f_threshold$$

Figure 7. Systems Architecture



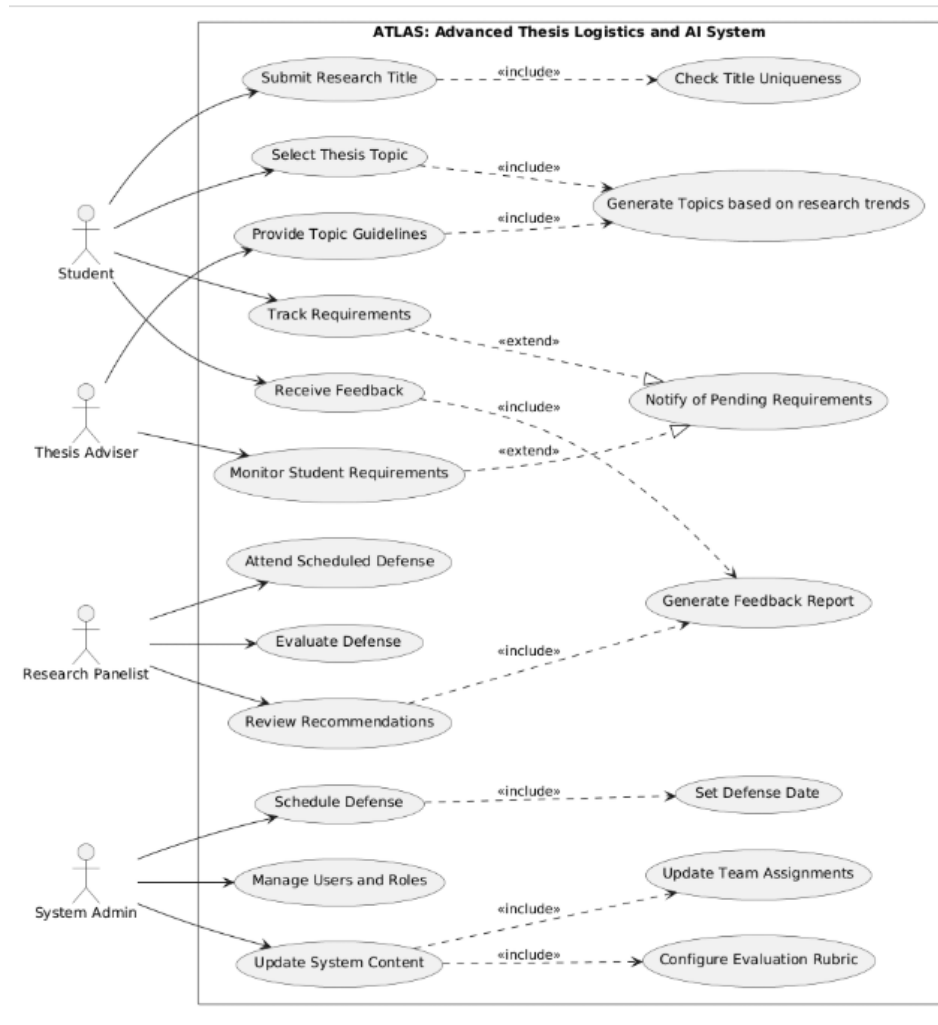
Object Modelling

This portion of the design of the application demonstrates the expected client of the application and the capacities that were accessible for the clients to utilize.

Use Case Diagram

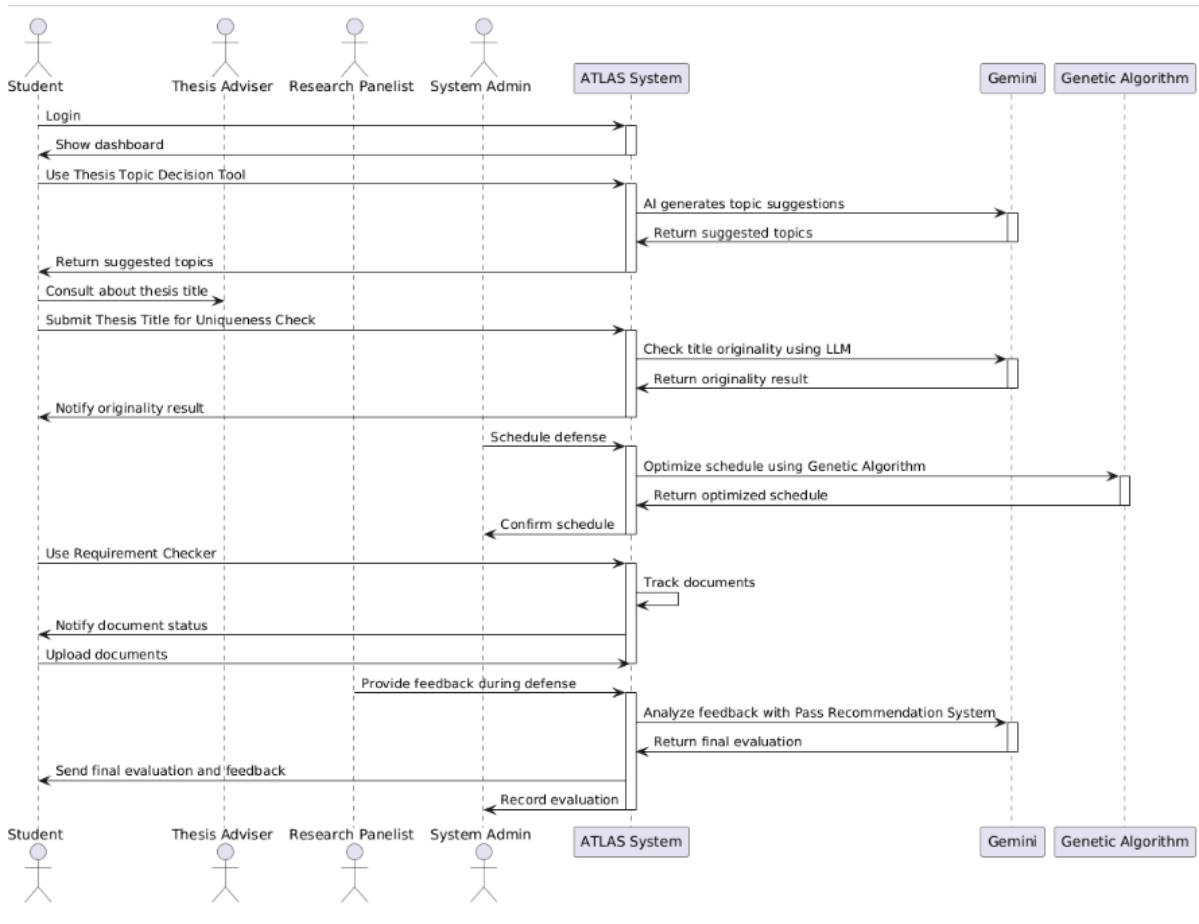
The ATLAS system streamlines thesis logistics by enabling students to select topics, submit titles, and track requirements, while advisers and panelists oversee evaluations and feedback. The system admin manages scheduling, user roles, and system content, ensuring an efficient and transparent thesis process.

Figure 8. Use Case Diagram



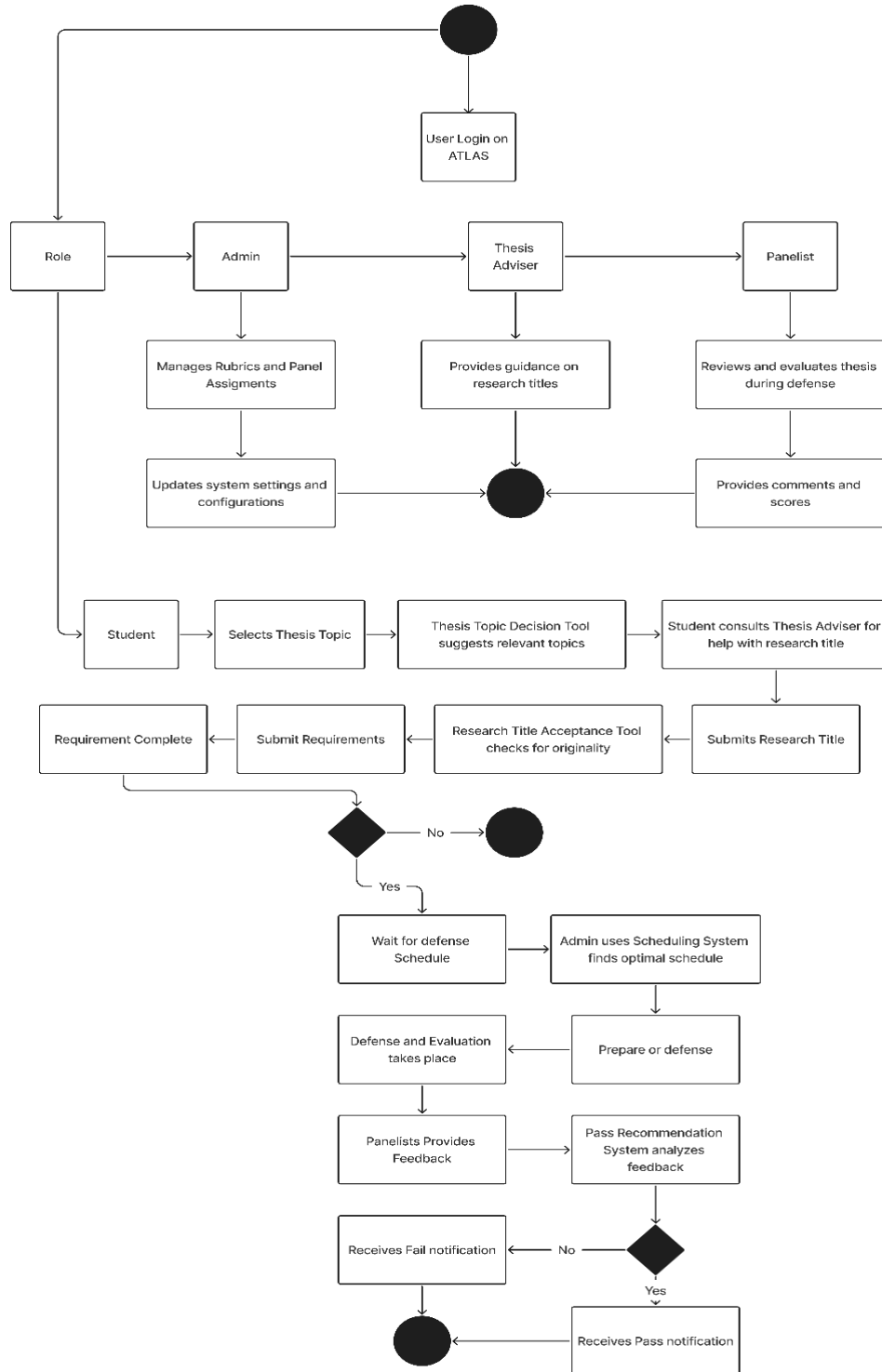
Students initiate the process by selecting a topic with assistance from the Suggest Relevant Topics feature, ensuring relevance. They then submit a research title, which is verified for originality using Check Title Uniqueness. Students also track their document submissions through Track Requirements, with optional reminders for incomplete tasks, and receive structured feedback after their defense through the Generate Feedback Report. Thesis Advisers guide students by refining

Figure 9. Sequence Diagram



The process starts with the student logging in to use the Thesis Topic Decision Tool, where AI generates tailored topic suggestions. After consulting their Thesis Adviser, the student submits the chosen title to the system for an AI-powered originality check. The system evaluates the title and notifies the student of the results. The System Admin schedules the defense using optimization algorithms, while the student uploads required documents, tracked by a Requirement Checker. During the defense, the Research Panelist provides feedback, which the system

Figure 10. Activity Diagram



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	Students begin by selecting a thesis topic and using the Thesis Topic Decision Tool for suggestions. They consult their thesis adviser for guidance on their research title before submitting it for originality verification via the Research Title Acceptance Tool. Afterward, students complete and submit all requirements. Once all requirements are fulfilled, students request a defense schedule, and the system's scheduling tool assigns an optimal time slot. During the defense, panelists review and evaluate the thesis, providing feedback and scores. The Pass Recommendation System analyzes this feedback to determine the final result, notifying the student of a pass or fail outcome. Admins manage rubrics, panel assignments, and system configurations, while thesis advisers guide students on research and provide feedback during the defense. Panelists focus on thesis evaluation and scoring. This process ensures efficient collaboration and streamlines thesis management.	
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Data Analysis

20 identify key challenges faced in the research process. A notable 65% of them have difficulty getting their research titles approved, and 70% struggle with selecting a suitable research topic. This emphasizes the necessity for AI tools such as the Research Title Acceptance Tool and Thesis Topic Decision Tool to improve these areas. What's more, 70% of students encounter issues with timely submissions of requirements, while 65% face confusion or delays regarding defense schedules. AI-driven solutions like a requirement tracker and an automated scheduling system could effectively resolve these challenges. When it comes to choosing research topics, 70% favor AI-powered suggestion tools, and 80% use online resources such as Google Scholar. To ensure the uniqueness of titles, 55% prefer using AI tools, although 50% also appreciate feedback from peers or mentors, indicating that a hybrid approach may be advantageous. In managing document submissions, 45% opt for a manual checklist, while 40% prefer an AI-driven requirement tracker and guidance from faculty. This suggests that integrating AI with traditional methods could offer the most flexibility. Regarding evaluation tools, 55% find an AI-powered pass recommendation system quite useful, and 75% view a centralized system for managing defense requirements as highly beneficial. Opinions on AI-generated performance evaluations are varied; 45% find them very useful, while 25% still prefer traditional methods, indicating that AI should complement human evaluation rather than replace it.

Ethical Considerations

In the development of ATLAS: Automated Research Scheduling Using Genetic Algorithm and Gemini Assisted Evaluation for LPU-Cavite COECSA Department, the following considerations are needed to ensure the protection of all stakeholders and the responsible use of AI technology.

1. Data Privacy and Confidentiality

- The proponents ensures that all the data/information gathered throughout the duration of this project will be handled with strict confidentiality. The proponents ensures that unauthorized users will not have access to this data. All participants will stay anonymous.

2. Informed Consent

- Before participating in this study, all participants are informed about the objectives and procedures involved. The proponents ensure that the participants fully understand, their rights and the voluntary nature of their involvement. Consent will be obtained through signed letters by the Dean of the department.

3. Intellectual Property and Plagiarism

- The system will respect and protect the intellectual property of both participants and the institution. Research titles, theses, and other academic content submitted by the institution and students will be safeguarded within

the system. The plagiarism detector will ensure originality and have no similar existing titles.

5. Cybersecurity and Protection from Harm

- ATLAS will be developed with strong cybersecurity protocols to prevent unauthorized access, data breaches, or system manipulation. Protection from harm also includes ensuring that the system does not expose participants to any form academic, emotional, or reputational harm through errors or misuse.

6. Bias and Fairness

- To address potential algorithmic bias in ATLAS, specific attention was given to ensure fairness in the Genetic Algorithm used for scheduling and Gemini LLM used in Thesis Topic Decision Tool, Research Title Acceptance Tool, and Pass Recommendation Tool. Actions were made to prevent any biases that might favor groups or lead to unfair outcomes.
- Transparency is also a guiding principle in this study. A clear documentation was provided on how the proponents gathered data, used the data in this study, and analyzed the data within ATLAS.

7. Accountability and Integrity

- The researchers and developers of ATLAS will maintain accountability throughout the system's design, deployment and operation. The research

team also ensures that ATLAS operates with the highest level of academic integrity, supporting fair evaluation practices.

8. Collaboration Ethics

- ATLAS encourages collaboration between students, faculty, and the administrative staff while upholding ethical standards in all interactions. Contributions made by all participants during the systems development will be acknowledged. The researchers and developers of ATLAS ensures that the system will be used as a tool for academic growth while promoting responsible conduct in research and academic evaluation.

These ethical considerations ensure that an IT thesis or capstone project is conducted responsibly, safeguarding the interests of individuals, society, and the broader tech community.

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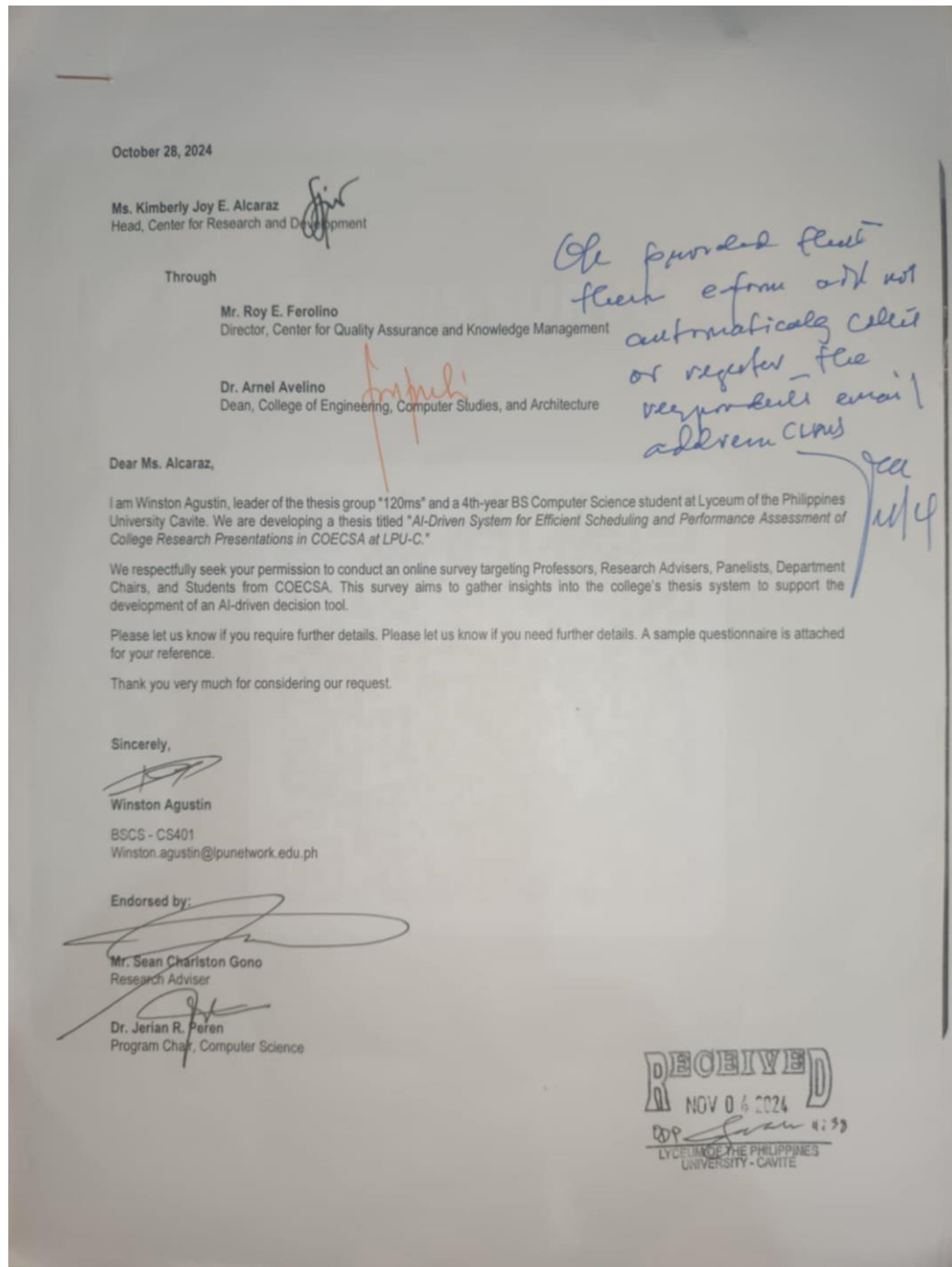
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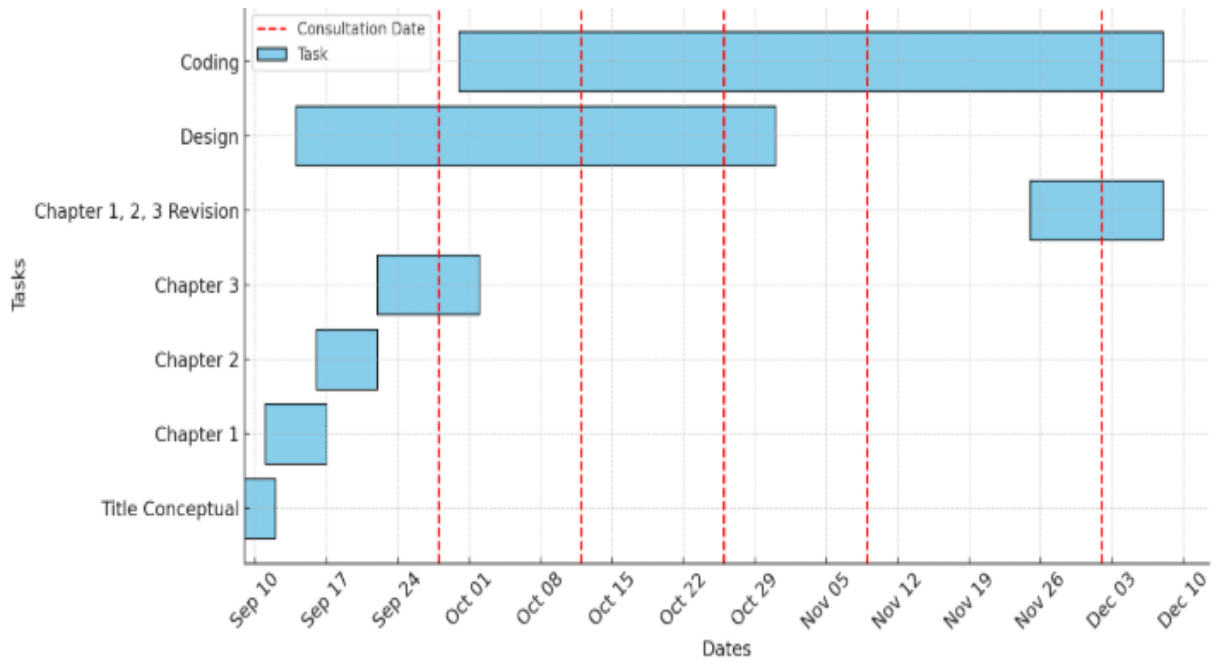
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APPENDICES

APPENDIX A LETTER OF REQUEST



**APPENDIX B
GANTT CHART**



**APPENDIX C
WIREFRAME**

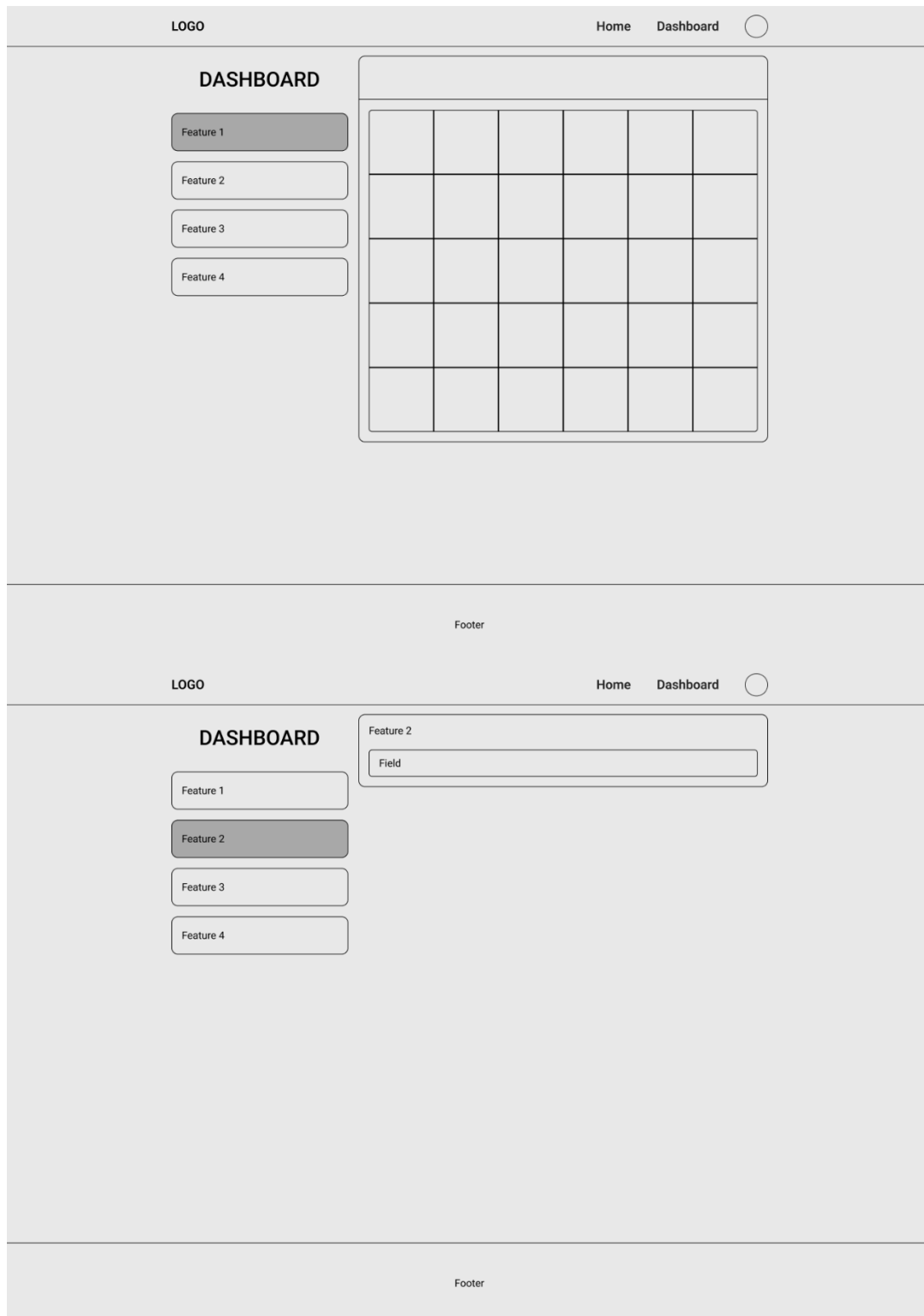
LOGO

Username

Password

[Forgot Password?](#)

Login



LOGO

HomeDashboard

DASHBOARD

Feature 1

Feature 2

Feature 3

Feature 4

Feature 3

Field

Field

Button

Footer

LOGO

HomeDashboard

DASHBOARD

Feature 1

Feature 2

Feature 3

Feature 4

Feature 4

Requirement
Details
Details
Details

Button

Requirement
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Details
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Requirement
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Dashboard feature 1

Dashboard feature 2

Dashboard feature 3

Dashboard feature 4

Dashboard feature 5

Dashboard feature 6

Dashboard feature 7

Dashboard feature 8

Dashboard feature 9

Dashboard feature 10

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Dashboard feature 1

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Dashboard feature 6

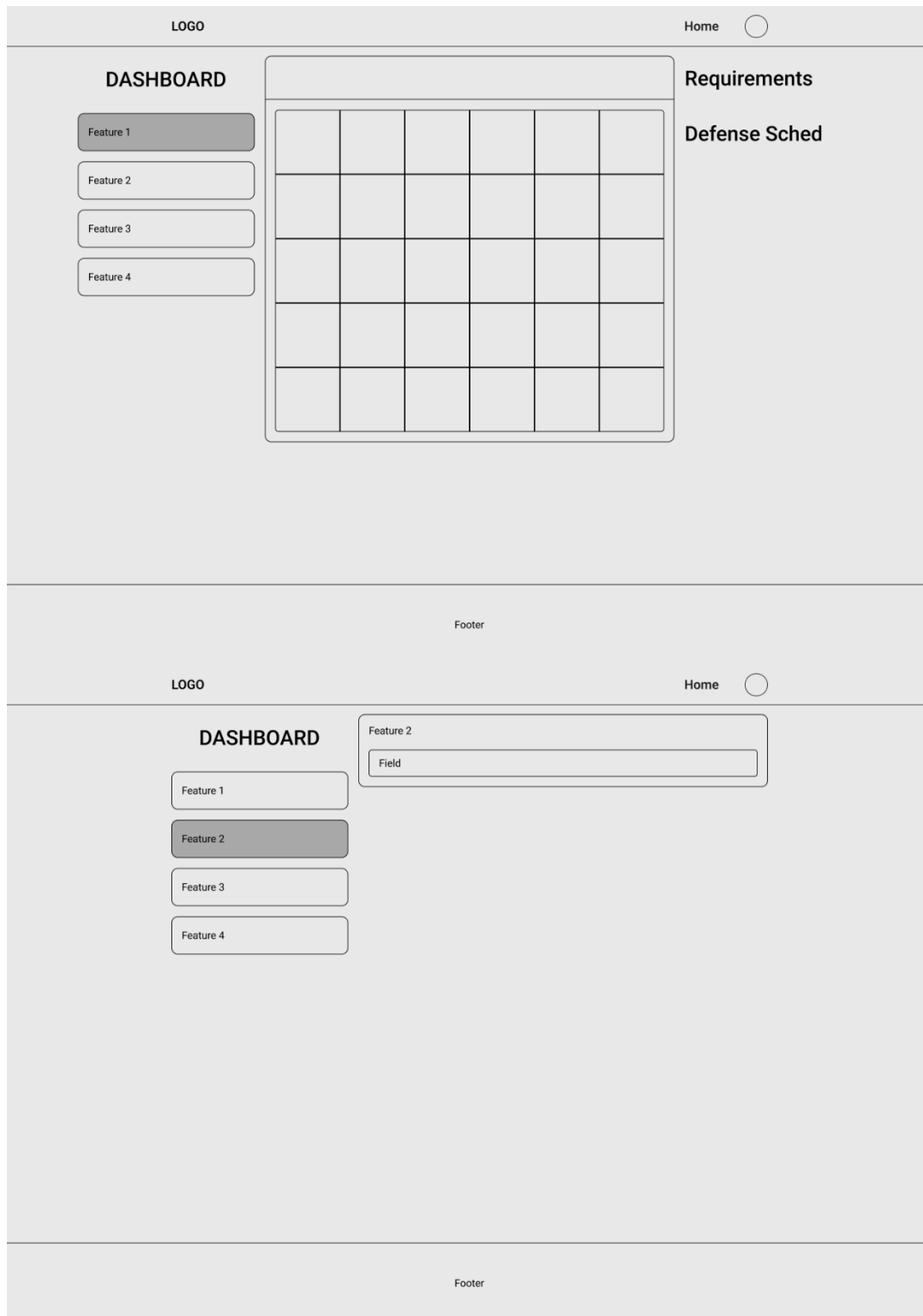
Dashboard feature 7

Dashboard feature 8

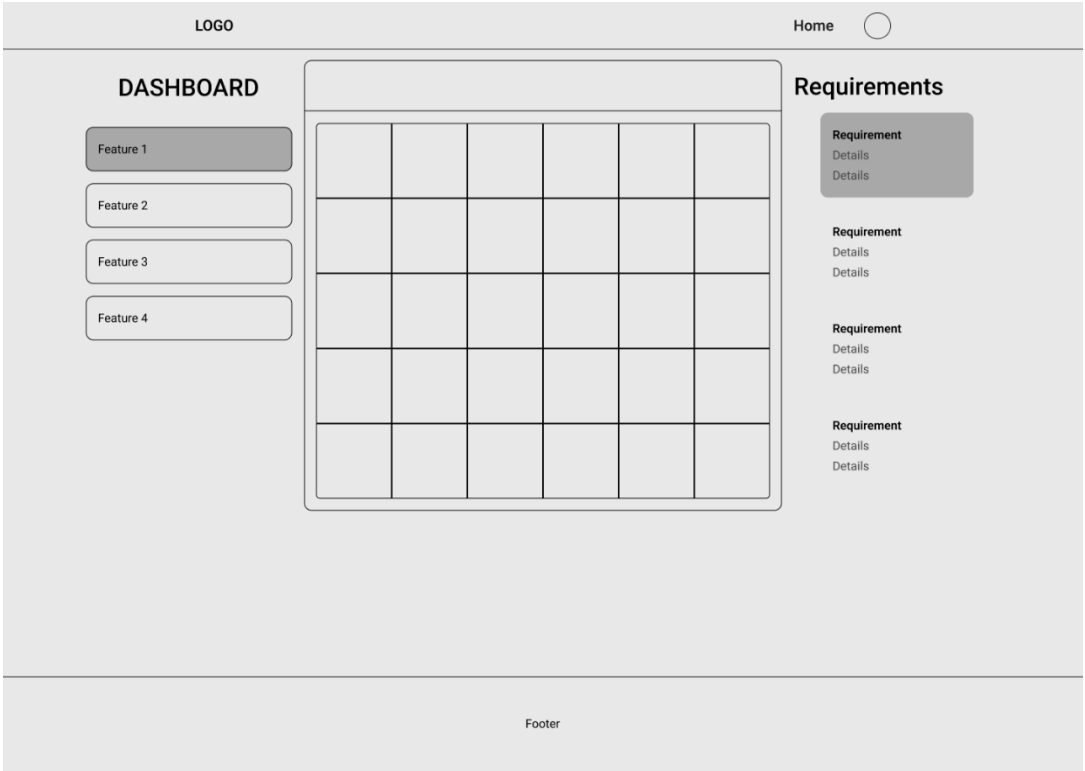
Dashboard feature 9

Dashboard feature 10

Footer







**APPENDIX D
PSEUDOCODE**

BEGIN

ENABLE error reporting and configure logging
SET content type to JSON

TRY

IMPORT necessary database and helper functions

IF database connection is not valid THEN
THROW exception "Database connection error"

IF request method is POST THEN
FETCH teams and panelists from database
PARSE input data (duration, rooms, timeSlots, days) from POST request

VALIDATE duration is a positive integer
IF invalid THEN
THROW exception "Invalid duration"

FETCH user schedules from database

RUN genetic algorithm with parameters:
teams, panelists, rooms, timeSlots, days, userSchedules, population size,
generations, mutation rate

IF saving schedule to database is successful THEN
PREPARE success response including:
- Initial population size
- Crossover and mutation counts
- Conflict counts and fitness scores per generation
- Population per generation

SAVE response to a JSON file
OUTPUT success response as JSON

ELSE
THROW exception "Failed to save schedule to database"

ELSE
THROW exception "Invalid request method"

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```
CATCH exceptions
  LOG error details
  OUTPUT error response as JSON
```

```
FUNCTION fetchTeams(pdo)
  EXECUTE SQL query to fetch teams with their advisers
  RETURN results
```

```
FUNCTION fetchPanelists(pdo)
  EXECUTE SQL query to fetch panelist IDs
  RETURN results
```

```
FUNCTION fetchUserSchedules(pdo)
  EXECUTE SQL query to fetch user schedules
  RETURN schedules organized by user ID
```

```
FUNCTION geneticAlgorithm(pdo, teams, panelists, rooms, timeSlots, days,
userSchedules, populationSize, generations, mutationRate)
  INITIALIZE population using `createInitialPopulation`
```

```
  FOR each generation
    CALCULATE fitness for each schedule in the population

    PERFORM selection on population
    REPLACE population with new schedules generated through crossover and
mutation
```

```
  SORT population by fitness DESCENDING
  RETURN the best schedule
```

```
FUNCTION createInitialPopulation(pdo, populationSize, teams, panelists, rooms,
timeSlots, days)
  FOR each individual in population
    CREATE DefenseSchedule object with randomly assigned panelists, rooms, times,
and days
  RETURN population
```

```
FUNCTION saveScheduleToDatabase(pdo, schedule)
  START database transaction
  FOR each defense in schedule
    INSERT defense details into the database
  COMMIT transaction
  RETURN true
```

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	<pre>FUNCTION DefenseSchedule(pdo, teams, panelists, rooms, timeSlots, days) INITIALIZE schedule with chromosomes for each team ASSIGN random panelists, rooms, times, and days FUNCTION hasConflicts(pdo, defense, userSchedules, all_defenses) CHECK conflicts with user schedules and overlapping times RETURN true if conflicts exist, false otherwise END</pre>	
	COLLEGE OF ENGINEERING, COMPUTER STUDIES, AND ARCHITECTURE	90